



香港科技大學(廣州)

THE HONG KONG UNIVERSITY OF SCIENCE
AND TECHNOLOGY (GUANGZHOU)

CNGF 5020 mini project

Investigating Agricultural Burning with Remote Sensing Data

----- **Bugs Team**



01. Core Task 1

What are the dominant spatiotemporal patterns of fire activity in the study region

02. Core Task 2

To what extent can the observed fire hotspots be attributed to the post-harvest burning of corn and wheat

03. Core Task 3

What are the long-term spatiotemporal trends of fire activity (2010–2019)

04. Core Task 4

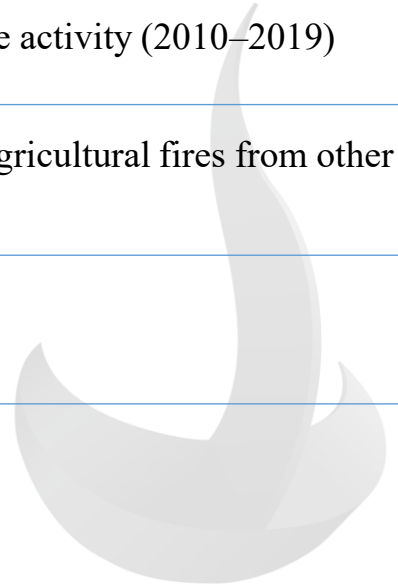
Are there distinct characteristics that differentiate agricultural fires from other Fires + Challenge 1: Intensity and Impact Analysis

05. Challenge 2

Methodological Limitations and Uncertainty

06. Challenge 3

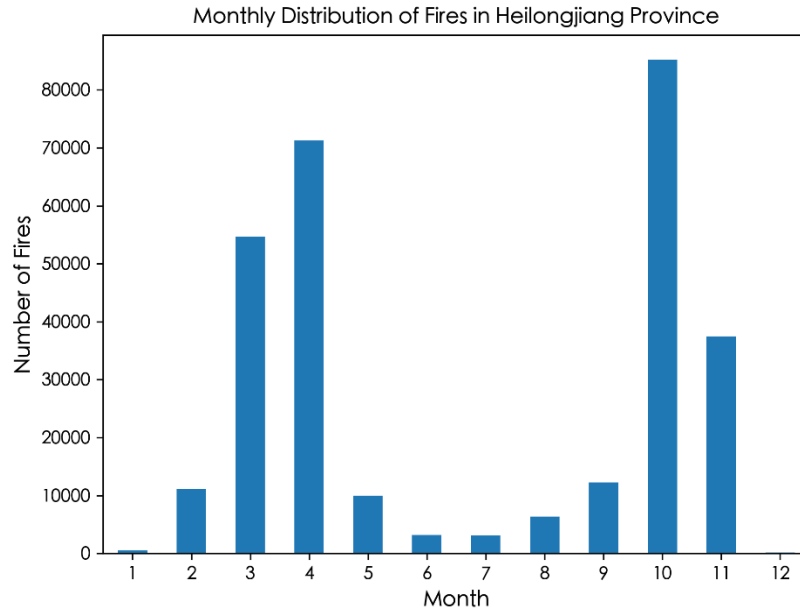
Meteorology & Pollution



Task 1



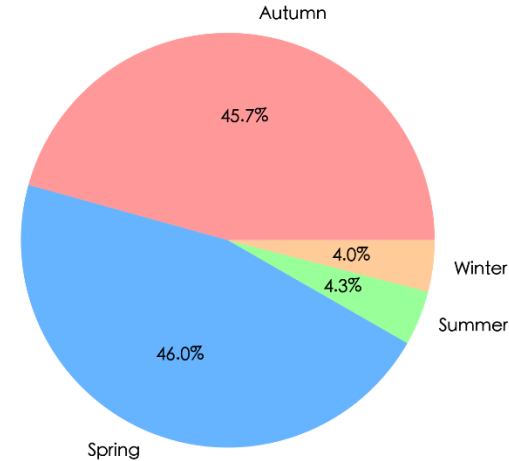
Time Distribution of Fires



Monthly Distribution

- Peak: October
- Months (January, June, July & December) see very few fires.

Seasonal Distribution of Fires in Heilongjiang Province



Seasonal Distribution

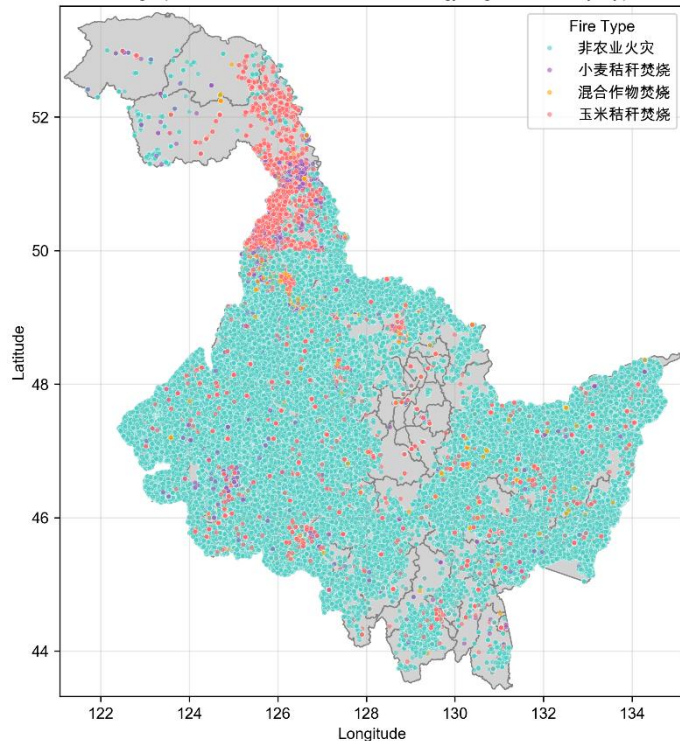
- Spring (46.0%) & Autumn (45.7%) account for most fires
- Summer (4.3%) & Winter (4.0%) have relatively low shares.

Spring: Mar-May
Summer: Jun-Aug
Autumn: Sep-Nov
Winter: Dec-Feb



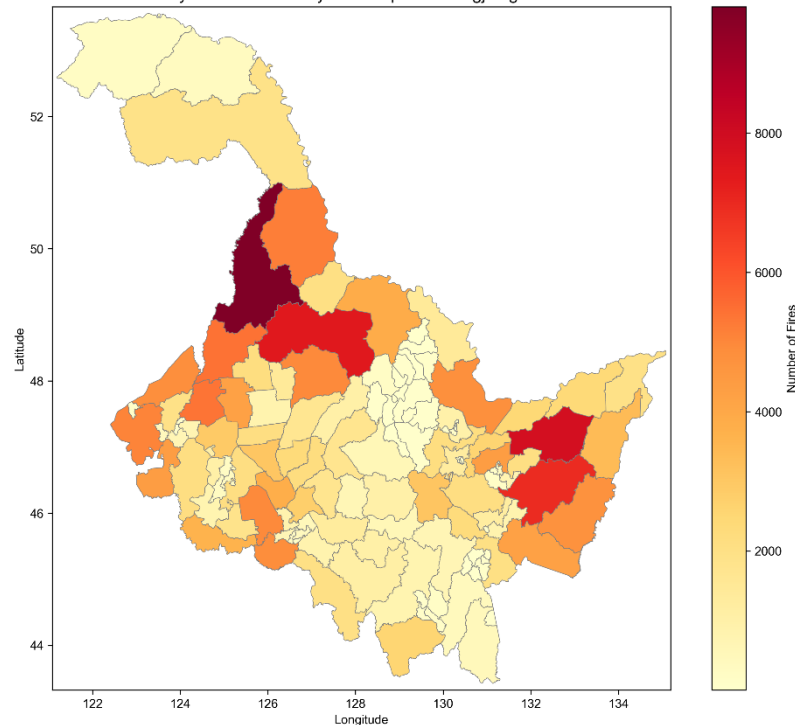
Geographic Distribution of Fires

Geographic Distribution of Fires in Heilongjiang Province by Type



Corn straw burning is concentrated in northern Heilongjiang, while non-agricultural fires are more widespread across the province.

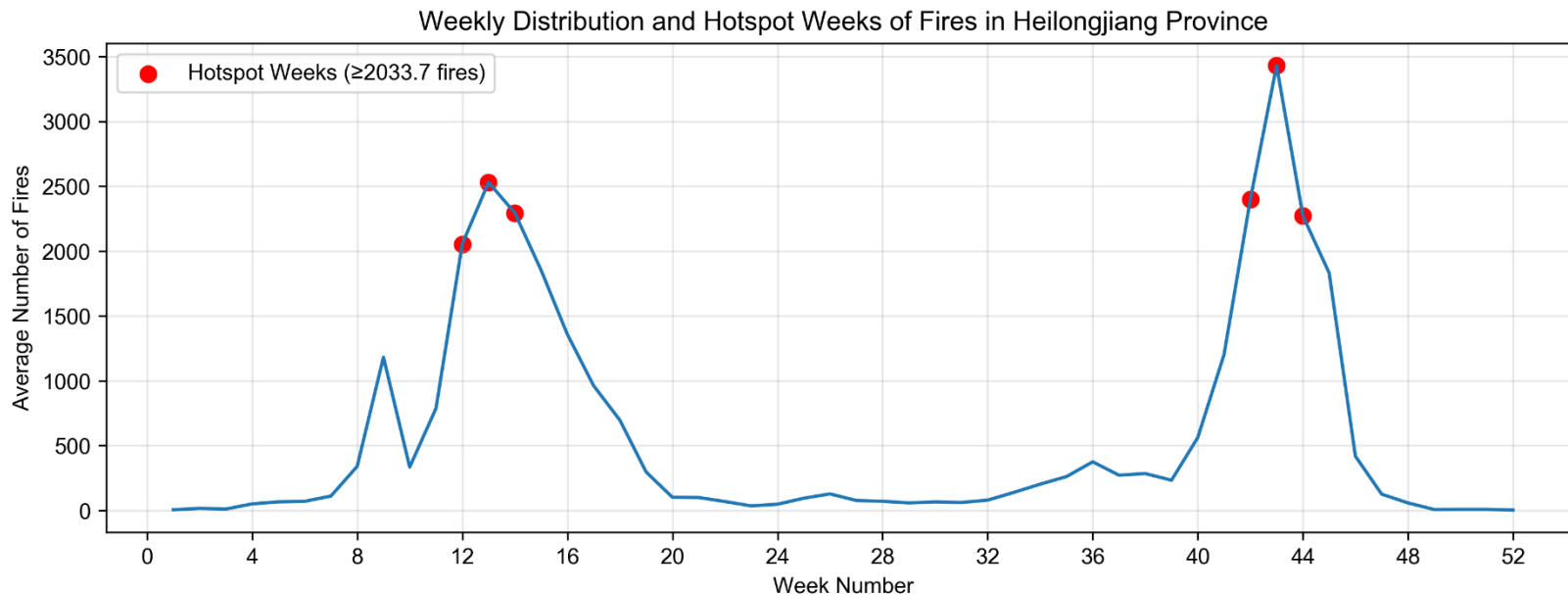
County-level Fire Density Heatmap in Heilongjiang Province



Dark red counties are fire “hotspots” with the highest fire density, guiding targeted fire prevention efforts.



Weekly Distribution and Hotspot Weeks



There are two clusters of hotspot weeks(Week14&42) in a year, which are key periods for fire control.



Task 2



Analysis of The Cause of The Fire

Method Design

- Step1: The maturity dates of crops at each location
 - ◆ Extract the position of each pixel point and the corresponding value of the grid point.
 - ◆ Convert the corresponding coordinates to latitude and longitude.
- Step2: Set the agricultural fire window period
 - ◆ Set the 5 days before and the 10 days after the maturity of each region's crops as the window period for burning crop straw. (-5, +10)



X坐标	Y坐标	成熟度日(DOY)
24476765	374743238.5	217
45918344	704489744.9	201
46322902	710708509.2	201
47536576	729404802	201
66146249	1015642959	193
66550807	1021860723	193
67966760	1043670398	193
67966760	1043554398	217
68573597	1052895544	201
69382713	1065323073	225
70191829	1077888601	193
70394109	1080987983	193
70394109	1080917983	209
70596388	1084107366	193
72214620	1108878423	217
74641969	1146342008	201
74641969	1146112008	209
74844248	1149447390	217

X坐标	Y坐标	成熟度日(DOY)
3736.398008	1657.565027	249
5089.309606	2246.918275	241
5089.309606	2246.907499	241
7149.263894	3156.268456	241
7149.263894	3156.25768	241
7149.263894	3156.246904	241
12965.60541	5728.140624	241
12965.60541	5728.129848	241
12965.60541	5728.119072	249
12965.60541	5728.108296	249
12965.60541	5728.09732	241
12965.60541	5728.084416	241
13086.7792	5781.02265	249
13086.7792	5781.011874	249
13086.7792	5781.001096	249
13207.95298	5834.562012	257
13329.12676	5888.521638	249
13329.12676	5888.133702	241
13329.12676	5888.122926	241
13450.30054	5941.759273	249
13450.30054	5941.748497	249



Analysis of The Cause of The Fire

Method Design

- Step3: Match the dates of each region's fire occurrences with the period of straw burning.

- Checking whether the area is a region where corn or wheat is cultivated.
- Checking whether the date of the fire occurred is within the burning window period.

latitu	longiti	lightn	scan	trac	acq_date	acq_time	satellite	instrument	onfide	versic	bright ti	frp	daynig	type	year	day_of_year	火灾类
43.5863	121.2319	300.8	1	1	2010年2月9日		454 Aqua	MODIS	42	6.2	271.3	7.6 D	0	2010		0	非农业火灾
43.5864	121.2446	314.3	1	1	2010年2月9日		454 Aqua	MODIS	75	6.2	271.7	17.0 D	0	2010		0	非农业火灾
43.5781	120.6539	309.1	1	1	2010年2月19日		506 Aqua	MODIS	68	6.2	280.5	9 D	0	2010		0	非农业火灾
43.4156	121.7671	307.5	1	1	2010年2月21日		454 Aqua	MODIS	65	6.2	274.1	10.2 D	0	2010		0	非农业火灾
43.4137	121.7546	303.3	1	1	2010年2月21日		454 Aqua	MODIS	52	6.2	274.5	7.6 D	0	2010		0	非农业火灾
43.683	123.3241	300.8	1.1	1	2010年2月23日		257 Terra	MODIS	41	6.2	275.4	6.7 D	0	2010		0	非农业火灾
44.9411	121.59	328.6	1.3	1.1	2010年2月28日		315 Terra	MODIS	86	6.2	273.4	44.7 D	0	2010		0	非农业火灾
44.9317	121.3888	311	1	1	2010年2月28日		500 Aqua	MODIS	71	6.2	275.8	12 D	0	2010		0	非农业火灾
43.2076	120.4746	307.9	1	1	2010年3月2日		304 Terra	MODIS	65	6.2	274.9	8.9 D	0	2010		0	非农业火灾
45.4896	122.9099	317	1.3	1.1	2010年3月9日		300 Terra	MODIS	77	6.2	277.3	25 D	0	2010		0	非农业火灾
43.6845	121.8011	301.1	1.2	1.1	2010年3月9日		300 Terra	MODIS	44	6.2	280.8	5.7 D	0	2010		0	非农业火灾
45.6438	122.417	301.2	1.2	1.1	2010年3月9日		300 Terra	MODIS	19	6.2	278.7	6.3 D	0	2010		0	非农业火灾
45.6628	122.4049	301.4	1	1	2010年3月9日		454 Aqua	MODIS	24	6.2	278.8	4.4 D	0	2010		0	非农业火灾
45.438	122.9442	311.7	1	1	2010年3月9日		454 Aqua	MODIS	72	6.2	278.2	12.4 D	0	2010		0	非农业火灾
45.7594	122.4395	300.3	1	1	2010年3月9日		454 Aqua	MODIS	35	6.2	277.5	4.7 D	0	2010		0	非农业火灾
45.6704	122.0155	302.6	1	1	2010年3月9日		454 Aqua	MODIS	53	6.2	284.6	5 D	0	2010		0	非农业火灾
45.5146	122.7926	301.1	2.3	1.5	2010年3月10日		214 Terra	MODIS	44	6.2	276.1	19.4 D	0	2010		0	非农业火灾
43.3605	120.4697	302.7	2	1.4	2010年3月10日		536 Aqua	MODIS	53	6.2	279.6	8.7 D	0	2010		0	非农业火灾
45.5668	122.3677	302	2.8	1.6	2010年3月10日		537 Aqua	MODIS	50	6.2	281.3	23.5 D	0	2010		0	非农业火灾
45.565	122.3722	304	2.8	1.6	2010年3月10日		537 Aqua	MODIS	58	6.2	281.2	27.1 D	0	2010		0	非农业火灾
47.3491	121.4216	300.1	1	1	2010年3月11日		256 Terra	MODIS	25	6.2	278.8	7.1 D	0	2010		0	非农业火灾
45.6054	125.6023	319.4	1	1	2012年7月8日		244 Terra	MODIS	66	6.2	304.2	8.9 D	0	2012		0	非农业火灾
47.6044	126.0806	323.6	1	1	2012年7月8日		244 Terra	MODIS	57	6.2	304.3	13.1 D	0	2012		0	非农业火灾
47.6024	126.0937	323.5	1	1	2012年7月8日		244 Terra	MODIS	41	6.2	304.2	12.9 D	0	2012		0	非农业火灾
45.7044	125.2646	322.7	1	1	2012年7月8日		244 Terra	MODIS	14	6.2	305.6	9.1 D	0	2012		0	非农业火灾
46.1854	126.8074	317.1	1	1	2012年7月8日		244 Terra	MODIS	55	6.2	301.2	9.7 D	0	2012		0	非农业火灾
46.0571	126.0796	315.8	1.2	1.1	2012年7月8日		244 Terra	MODIS	46	6.2	301	10.7 D	0	2012		0	非农业火灾
44.9145	127.1579	321.5	1.1	1	2012年7月8日		244 Terra	MODIS	35	6.2	304.9	11.6 D	0	2012		0	非农业火灾
47.1442	127.3231	314.4	1.1	1	2012年7月8日		244 Terra	MODIS	62	6.2	299.9	7.3 D	0	2012		0	非农业火灾
46.5292	125.0218	327.4	1	1	2012年7月8日		244 Terra	MODIS	70	6.2	307.1	12.3 D	0	2012		0	非农业火灾
46.056	125.9338	320.2	1	1	2012年7月8日		244 Terra	MODIS	41	6.2	301.3	8.6 D	0	2012		0	非农业火灾
47.2471	127.1057	321.5	1	1	2012年7月8日		244 Terra	MODIS	57	6.2	303.2	10.6 D	0	2012		0	非农业火灾
42.7149	125.2105	320.5	1	1	2012年7月8日		245 Terra	MODIS	74	6.2	300.8	8.9 D	0	2012		0	非农业火灾
43.9218	125.2047	319.8	1	1	2012年7月8日		245 Terra	MODIS	60	6.2	301.6	8.4 D	0	2012		0	非农业火灾
43.6626	126.4987	321.6	1.1	1	2012年7月8日		427 Aqua	MODIS	59	6.2	304.1	13.1 D	0	2012		0	非农业火灾
45.194	127.8205	316.7	1	1	2012年7月8日		427 Aqua	MODIS	57	6.2	298.8	6.8 D	0	2012		0	非农业火灾
45.1859	127.8331	320.4	1	1	2012年7月8日		427 Aqua	MODIS	71	6.2	300.8	10.5 D	0	2012		0	非农业火灾
43.6534	126.5018	319.2	1.1	1	2012年7月8日		427 Aqua	MODIS	41	6.2	301.9	10.1 D	0	2012		0	非农业火灾
44.1117	129.2086	325.3	1	1	2012年7月8日		427 Aqua	MODIS	81	6.2	307.8	12.5 D	0	2012		0	非农业火灾
44.0114	126.3299	320.2	1.1	1	2012年7月8日		427 Aqua	MODIS	68	6.2	304.2	10.2 D	0	2012		0	非农业火灾
44.1525	125.8296	320.1	1.1	1.1	2012年7月8日		427 Aqua	MODIS	42	6.2	304.3	11.2 D	0	2012		0	非农业火灾
43.9681	125.2622	319.9	1.2	1.1	2012年7月8日		427 Aqua	MODIS	60	6.2	303.2	9.9 D	0	2012		0	非农业火灾
46.6034	131.0937	311.1	1.1	1.1	2012年7月8日		427 Aqua	MODIS	37	6.2	290.3	5 D	0	2012		0	非农业火灾
45.7216	125.4105	315.9	1.1	1.1	2012年7月8日		427 Aqua	MODIS	33	6.2	306.1	8.6 D	0	2012		0	非农业火灾

- Step4: Attribution analysis and fire classification

year	小麦秸秆焚烧	小麦秸秆焚烧占比	混合作物焚烧	混合作物焚烧占比	玉米秸秆焚烧	玉米秸秆焚烧占比	非农业火灾	非农业占比	总计
2010	480	0.029305818	157	0.009585445	4040	0.246657305	11702	0.714451432	16379
2011	1221	0.042573222	259	0.009030683	5633	0.196408647	21567	0.751987448	28680
2012	789	0.060997294	278	0.021492076	1076	0.083185157	10792	0.834325474	12935
2013	390	0.019277347	79	0.003904898	3439	0.169986654	16323	0.806831101	20231
2014	742	0.015695731	331	0.007001735	3627	0.076722934	42574	0.9005796	47274
2015	753	0.015913606	280	0.00591741	2075	0.043852234	44210	0.93431675	47318
2016	1228	0.036820485	642	0.019249798	1049	0.03145363	30432	0.912476388	33351
2017	781	0.015848536	201	0.004078817	2427	0.049250188	165870	0.8082246	49279
2018	817	0.042404111	253	0.013131261	1813	0.094098718	44374	0.850365911	19267
2019	439	0.017696618	155	0.006248236	1426	0.057483775	22787	0.918571371	24807
总计	7640	0.025507393	2635	0.00879738	26605	0.088825158	262641	0.876870069	299521

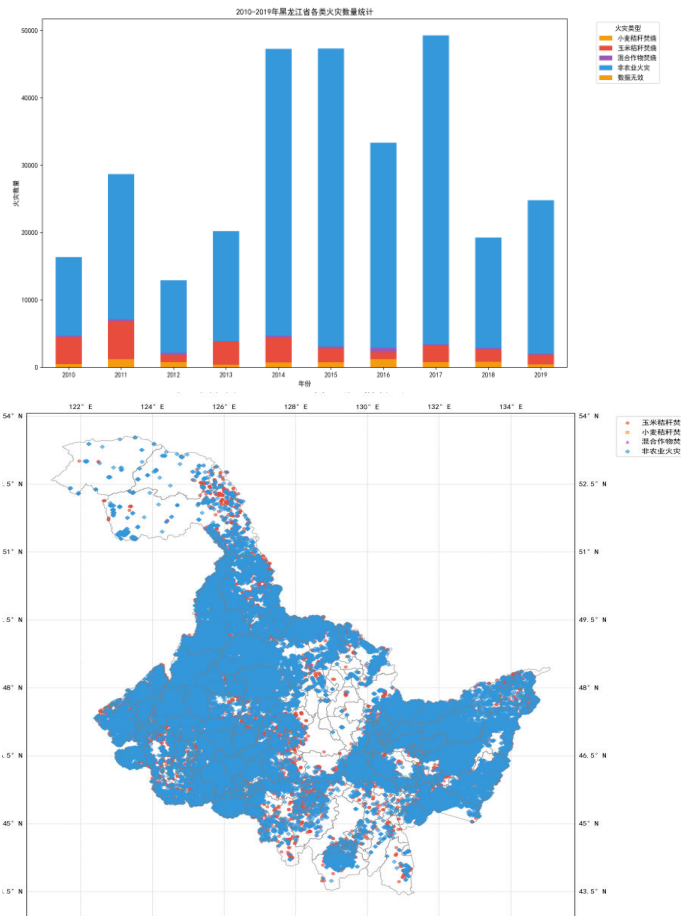


Analysis of The Cause of The Fire

Result Analysis and Visualization

Type of fire	The proportion of fires	Interannual fluctuation range	Interannual volatility
Corn firing	8.89%	3.42%-24.67%	0.073
Wheat firing	2.56%	1.61%-6.21%	0.015
Mixed firing	0.88%	0.03%-2.14%	0.006
Non agricultural firing	87.69%	71.44%-93.43%	0.078

- Conclusion 1: We classify 12.3% of the fires as possible agricultural burning.
- Conclusion 2: Corn stalk burning accounts for the highest proportion among all agricultural fires.

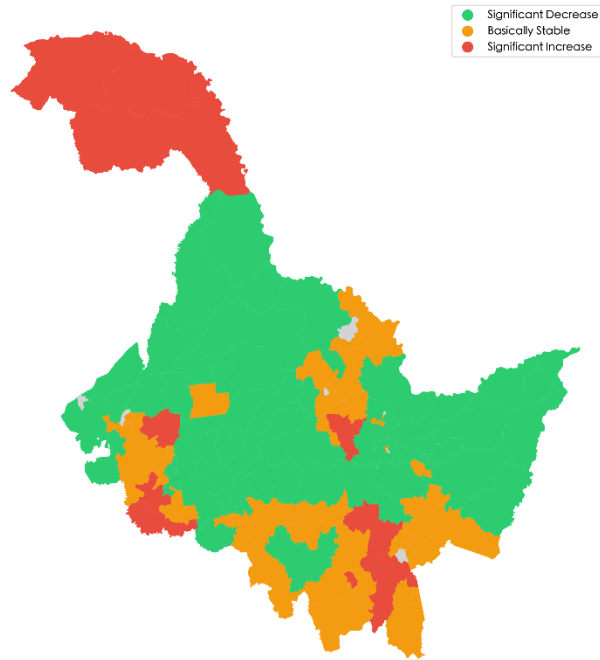


Task 3

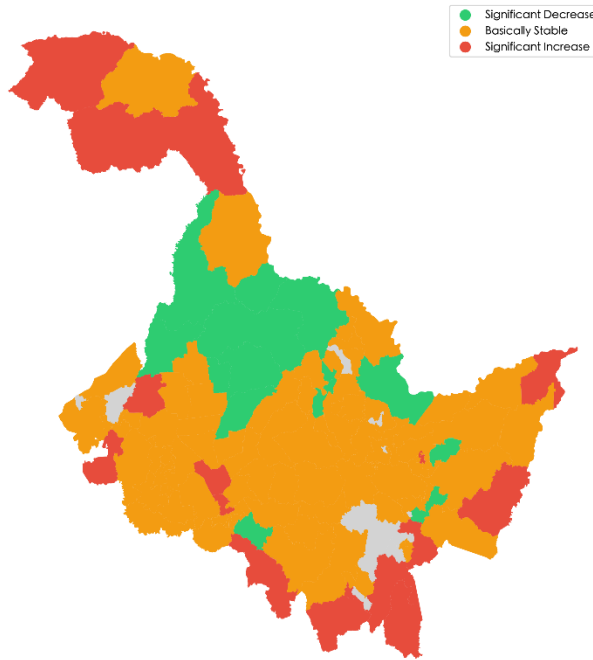


County-level Interannual Trend of Fires (2010–2019)

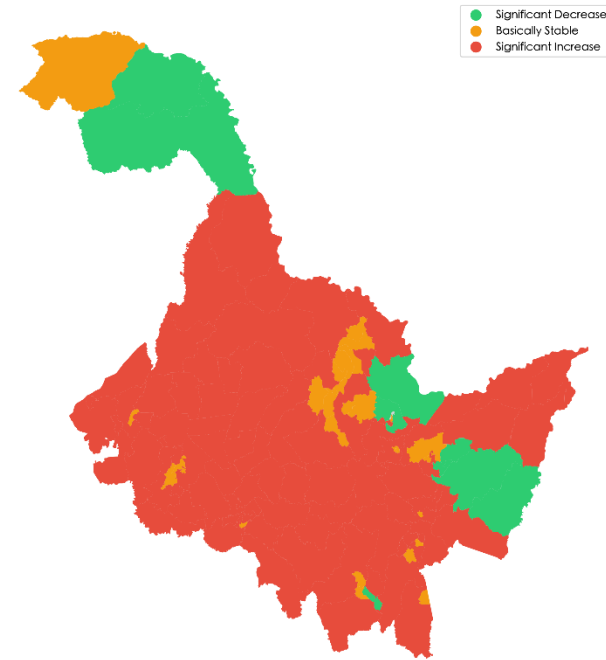
Heilongjiang County-level 玉米秸秆焚烧 Interannual Trend (2010-2019)



Heilongjiang County-level 小麦秸秆焚烧 Interannual Trend (2010-2019)

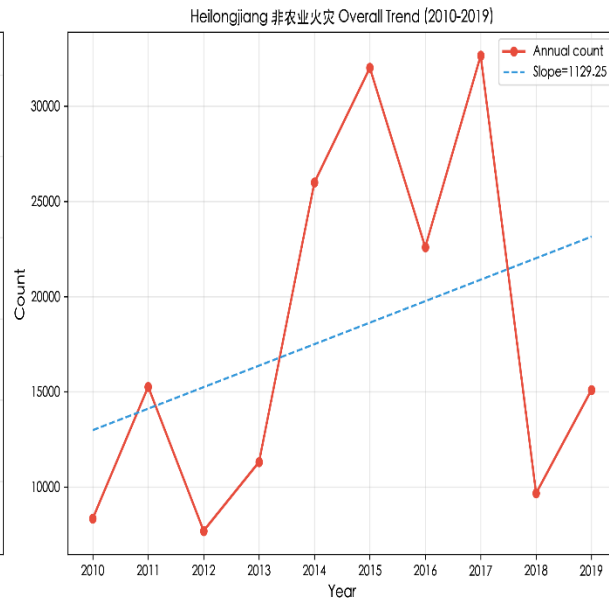
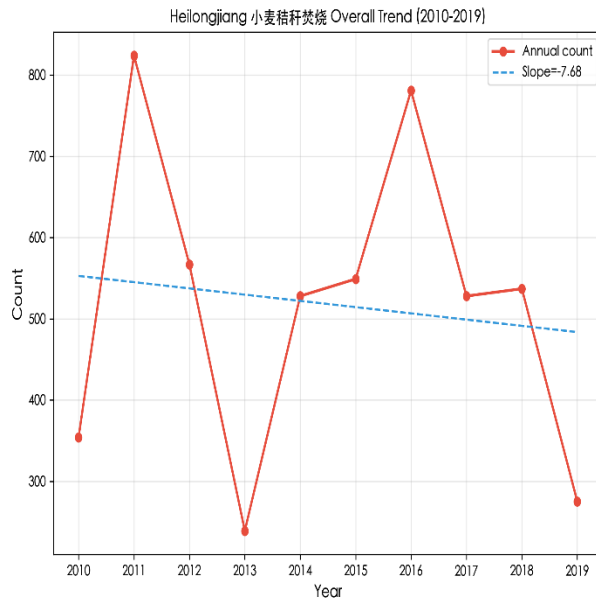
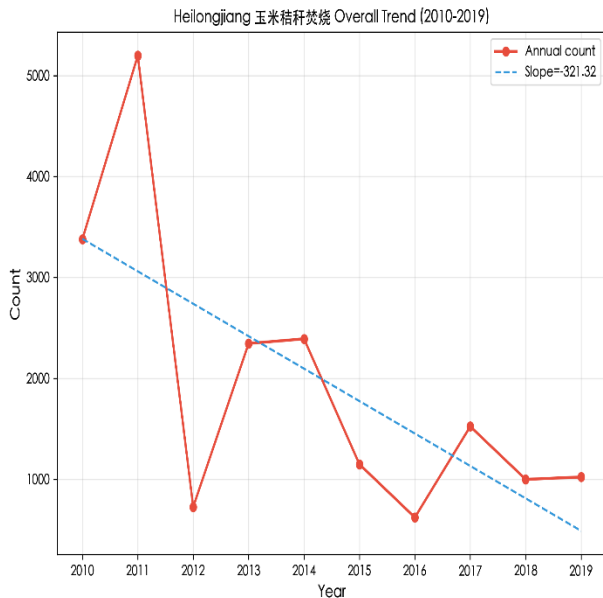


Heilongjiang County-level 非农业火灾 Interannual Trend (2010-2019)



- Corn Straw Burning: Most counties show a significant decrease, indicating effective control measures.
- Wheat Straw Burning: Trends are mixed, with counties showing decrease, stability, or increase.
- Non-agricultural Fires: Most counties show a significant increase, requiring further investigation into driving factors.

Overall Trend and Linear Fitting (2010–2019)



- Corn Straw Burning: The linear slope is -321.32, indicating a significant decrease over the decade.
- Wheat Straw Burning: The slope is -7.68, showing a slight downward trend.
- Non-agricultural Fires: The slope is 1129.25, reflecting a significant increase year by year.

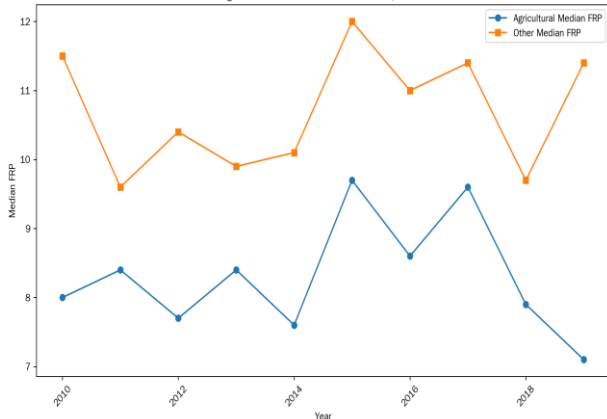


Task 4



Comprehensive Analysis of Agricultural vs Non-agricultural Fires

Agricultural vs Other Median FRP Comparison

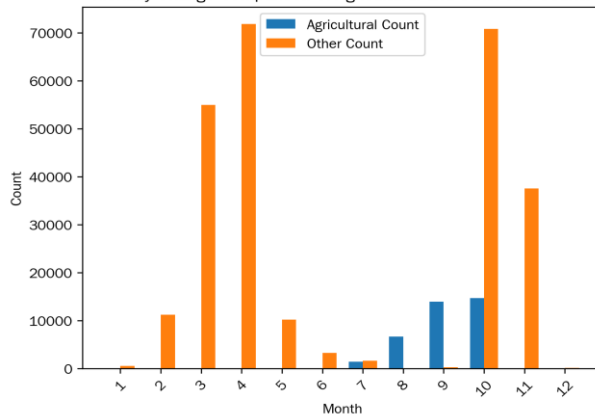


Fire Intensity

Agricultural fires: lower median FRP (~8–9), mostly low-intensity, small-scale burns.

Other fires: higher FRP (median 10.8, max 1824), often larger and more intense events.

Monthly Change Comparison of Agricultural and Other Fire Counts

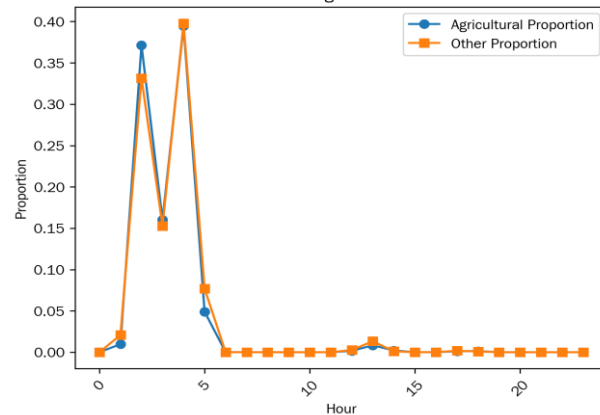


Monthly Variation

Agricultural fires: strongly concentrated in July–October, matching harvest and residue-burning seasons.

Other fires: show a bimodal pattern, peaking in spring (Mar–Apr) and autumn (Oct), indicating multiple driving factors.

Time Distribution of Agricultural and Other Fires



Hourly Distribution

Agricultural fires: mainly occur at night (2–5 a.m.), peaking around 4 a.m., reflecting nighttime burning habits.

Other fires: spread more evenly across hours, showing no clear temporal concentration.

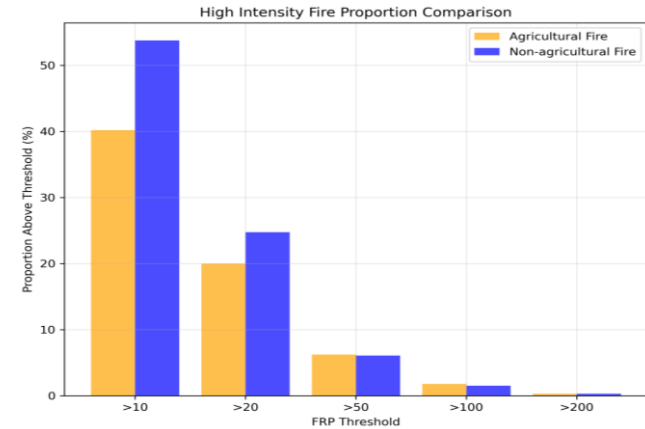


Challenge 1



FRP Analysis of Agricultural vs Non-agricultural Fires

	Average FRP	Median FRP	High intensity ratio
wheat	14.28	8.80	3.5%
maize	17.61	8.00	3.9%
other	18.42	10.80	6.1%



Agricultural Fires:

- Generally low-intensity burns with smaller FRP values.
- Often numerous but less impactful in terms of total energy release.
- Some maize fires show relatively high FRP, indicating occasional intense burnin

Other / Wildfires:

- Have higher average and peak FRP, showing greater intensity.
- Less frequent but more destructive, requiring more firefighting resources.
- Contribute significantly to environmental and ecological damage.



Challenge 2



Methodological Limitations & Uncertainties

Core Classification Method:

Estimates agricultural burning via spatial overlap with farmland and temporal proximity to harvest (30 days post-crop maturity) — a reasonable baseline but with key uncertainties.

Underestimation of agricultural fire points:

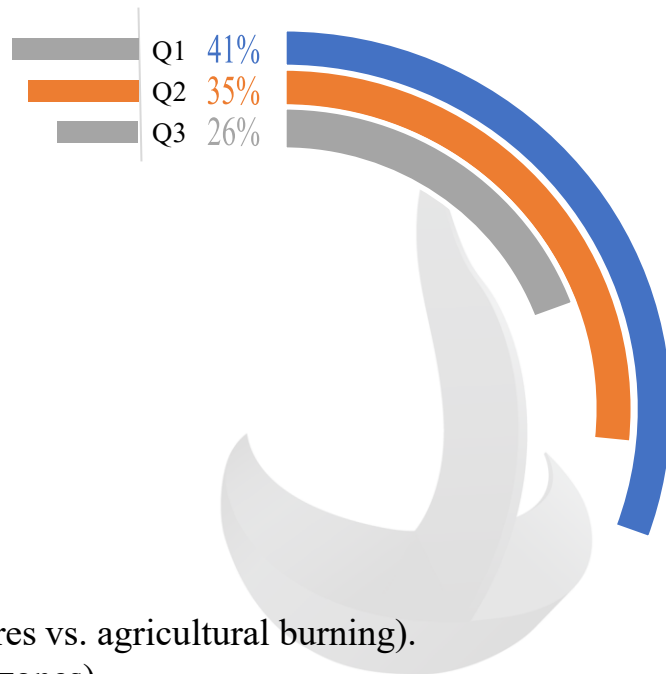
40% Caused by MODIS miss detection of small-scale fires and 1 km map failing to capture fragmented farmland.

Overestimation of agricultural fire points:

35% Caused by misjudgment of mixed pixels and grassland fires being incorrectly included.

Temporal mismatch:

25% Caused by static phenological data and the fixed 30-day window.



Improvement Strategies

High-res LU/LC: Globeland30 (30m) / Sentinel-2

VIIRS Nighttime Lights: Exclude urban/industrial fires.

Meteorological Data: Soil moisture/wind/temp (distinguish wildfires vs. agricultural burning).

Road Networks: Proxy for human activity (refine agricultural fire zones).



Challenge 3



Data Source:

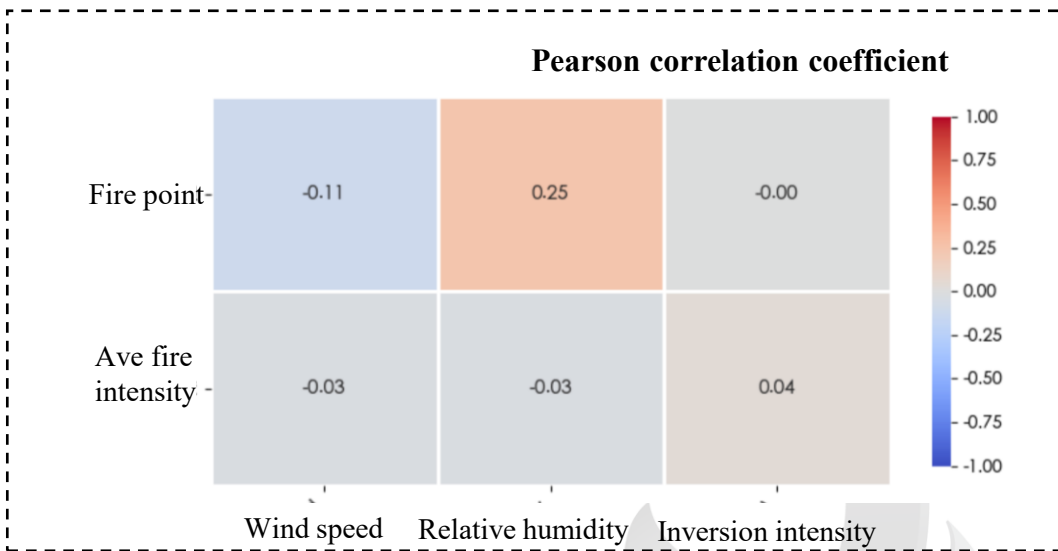
1. Meteorology

- **Wind speed(m/s)**-ERA5-Land hourly data from 1950 to present
- **Relative humidity(%)**-Agrometeorological indicators from 1979 to present derived from reanalysis
- **Inversion intensity (°C/100m)**-ERA5 post-processed daily statistics on pressure levels from 1940 to present

2. Pollution

- **PM_{2.5}($\mu\text{g}/\text{m}^3$)**-Wei, J., Li, Z. (2024). ChinaHighPM2.5: High-resolution and High-quality Ground-level PM_{2.5} Data set for China.

1. Monitoring of Straw Burning and Analysis of Environmental Impacts under Meteorology



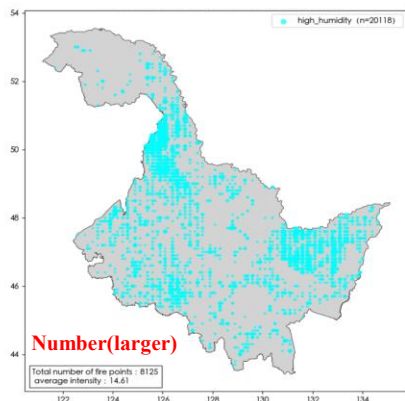
Fire point monitoring:

- Positive(+): relative humidity
- Negative(-): wind speed

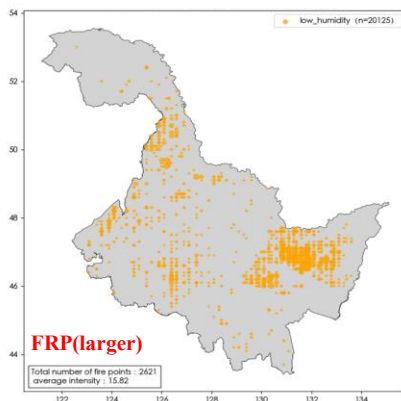
Average fire intensity:

- Positive(+): inversion intensity
- Negative(-) wind speed, relative humidity

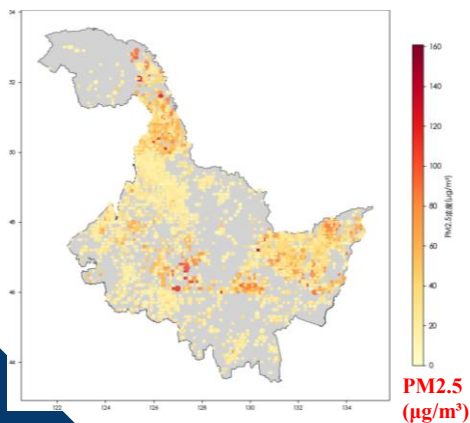
Fire point distribution (**High humidity**)



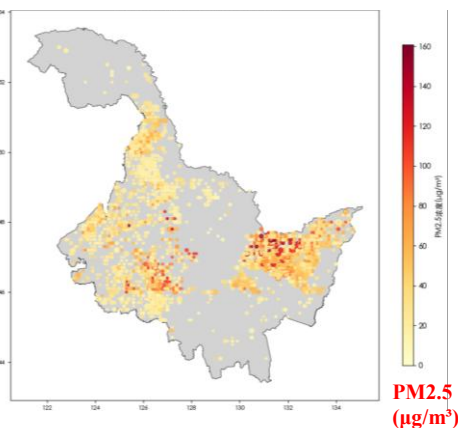
Fire point distribution (**Low humidity**)



Fire point and **PM2.5** distribution (**low wind speed**)

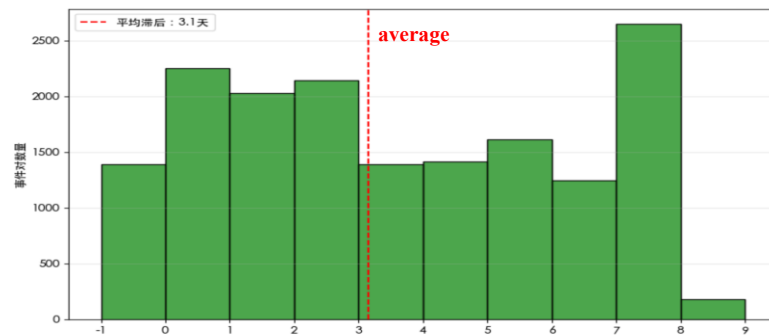


Fire point and **PM2.5** distribution (**high wind speed**)

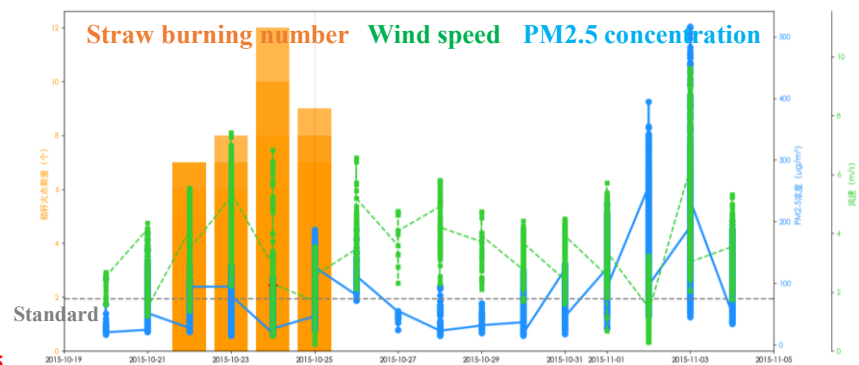


2. Correlation Analysis Between Pollution Peaks and Timing of Straw Burning

Distribution of Time Lags from 2010 to 2019



Time lag between the onset of burning and the **peak PM2.5** (day)



Correlation Analysis Between **Typical Straw Burning** in 2015 and **PM2.5** Peak Levels





Thank You
Q & A